

CS-127 · OOP LAB PROJECT

SLIDEDECK

Chess Game — Desktop Application

C# · .NET 10 · Windows Forms · GDI+

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Project Overview

3,978

Lines of Code

Across 10 source files

10

Source Files

Zero NuGet dependencies

4

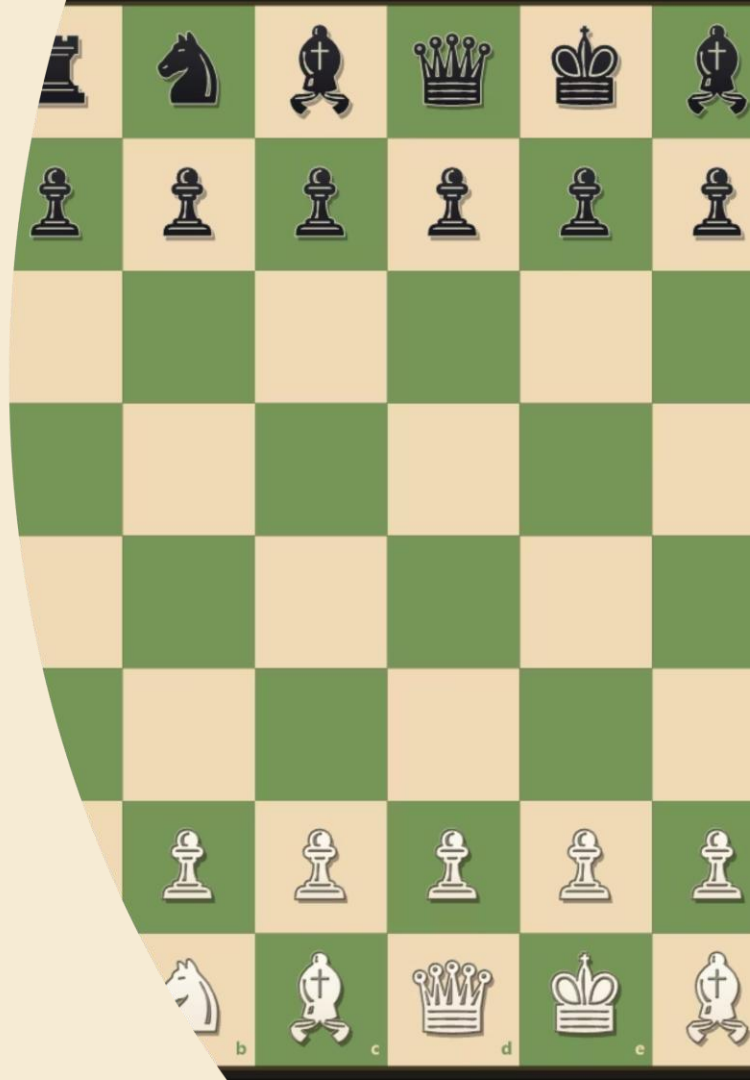
Board Themes

Forest, Ocean, Walnut, Carbon

3

AI Levels

Easy, Medium, Hard



Core Features



Two Game Modes

Player vs AI · Player vs Player (local)



Full Chess Rules

Castling, en passant, pawn promotion,
check/checkmate/stalemate



AI Opponent

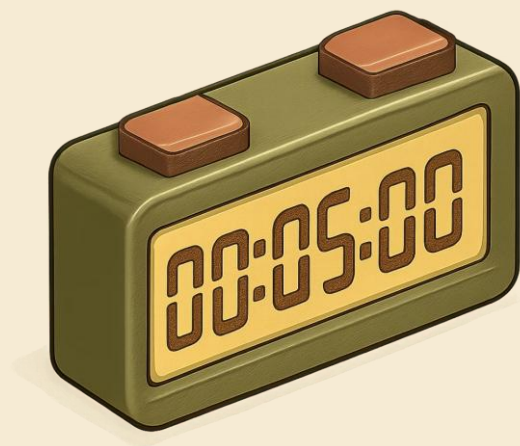
Minimax + Alpha-Beta pruning · Piece-Square Tables · Async
via Task.Run()



Algebraic Notation

All moves recorded with captures, castling, promotions,
check & checkmate symbols

Clock, History & Captures



Chess Clock

Blitz (5 min), Rapid (10 min), Long (30 min) — red alert below 10 sec, 100ms precision

Move History

Scrollable table, gold-highlighted latest move

Captured Pieces

Sorted by value with material advantage indicator

Comparative Study

Platform	Offline	AI Algorithm	Multiplayer	Clock	UI	Dependencies	Platform
Our App	✔ Yes	Minimax	Local PvP	3 Presets	GDI+	Zero	Windows
Chess.com	✘ No	Stockfish	Online PvP	✔ Yes	Web/Canvas	Required	Browser
Lichess	✘ No	Stockfish	Online PvP	✔ Yes	Web/SVG	Required	Browser
Python-Chess	✔ Yes	Basic	✘ None	✘ None	None	Required	Console

Our Unique Advantage

Offline play, zero external dependencies, and a custom GDI+ rendered UI — not simultaneously available in any compared alternative.

Class Architecture

01

Program.cs (16 lines)

Entry point

02

MenuForm.cs (704 lines)

Main menu, settings, game launch

03

ChessForm.cs (911 lines)

Game window — orchestrates all gameplay

04

ChessEngine.cs (835 lines)

Core logic — move validation & rules

05

ChessAI.cs (316 lines)

Minimax with Alpha-Beta pruning

06

ChessBoardControl.cs (669 lines)

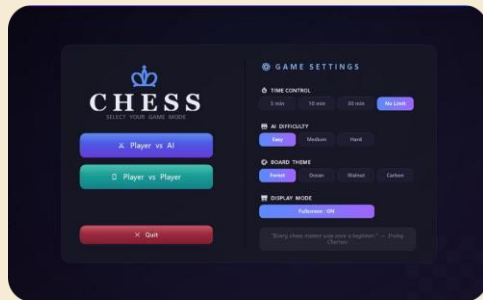
Custom GDI+ rendered interactive board

07

ChessClock.cs (211 lines)

Dual-timer countdown system

Screen Descriptions



Main Menu

Game mode select, AI difficulty, timer presets, board themes, fullscreen toggle



Game Interface

8x8 board with algebraic labels, sidebar with AI status, Undo / Reset / Flip / Menu



Endgame

Checkmate / stalemate detection with Play Again option

Technology Stack



Language & Framework

C# 13 · .NET 10.0 · MSBuild /
dotnet CLI · Visual Studio
2022



UI Framework

Windows Forms (WinForms)
with GDI+ custom rendering
— zero web dependencies



AI Algorithm

Minimax + Alpha-Beta Pruning + Piece-Square Tables (PST) for all 6
piece types



OOP Principles Applied



Production-Quality Desktop App

This project demonstrates comprehensive application of **Encapsulation, Inheritance, Polymorphism**, and **four design patterns** — within a fully functional, offline-capable chess application with zero external dependencies.

📄 3,978 lines · 10 files · 0 dependencies